Collaborative Machine Learning Project Using Jupyter Notebook

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1. Objectives

This project is intended to provide practical experience with machine learning techniques, covering all steps from data preprocessing to model evaluation and interpretation. You will select a dataset, conduct exploratory data analysis, prepare data for modeling, build and fine-tune machine learning models using two different approaches, and evaluate their performance. Working in pairs will foster collaboration and allow you to benefit from shared insights and problem-solving.

2. Project Description

In pairs, you will complete a machine learning project using Jupyter Notebook as the primary development environment. The project should include the following components:

2.1 Team Formation:

- Form a team of two students to work collaboratively throughout the project.
- Take advantage of pair programming and collaborative brainstorming to approach problem-solving from multiple perspectives.

2.2 Data Selection:

- Choose a dataset that both team members find interesting. It can be from fields like finance, healthcare, education, social media, or technology.
- The dataset should have sufficient complexity to support insightful machine learning analyses. Consider datasets from:
  - UCI Machine Learning Repository: A large collection of databases used widely in the machine learning community.
  - Kaggle Datasets: A platform with a variety of datasets for different domains.
  - Google Dataset Search: A tool to discover datasets available online.
  - Data.gov: A source of government datasets across various sectors.
  - The datasets proposed on our learning platform.
- Ensure compliance with data usage rights and confirm that the dataset does not contain sensitive or private information, unless anonymized.

2.3 Data Preprocessing:

- Address any issues with missing values, duplicates, and outliers.
- Apply feature engineering and scale data as needed to optimize model performance.

2.4 Exploratory Data Analysis (EDA):

- Use visualizations and statistical summaries to explore data distributions and relationships between variables. Employ visual tools like histograms, scatter plots, and heatmaps.
- Summarize the key insights gathered from the EDA, which will guide your model selection.

2.5 Model Selection:

- Select at least two different machine learning algorithms, focusing on diversity (e.g., one supervised and one unsupervised, or one regression and one classification).
- Justify your choice of algorithms based on the problem type and findings from the EDA.

2.6 Model Training and Tuning:

- Split the dataset into training and testing subsets (e.g., 80% for training, 20% for testing).
- Use cross-validation techniques to fine-tune model parameters and prevent overfitting.

2.7 Model Evaluation:

- Evaluate the models using appropriate metrics (e.g., accuracy, precision, recall, F1-score for classification, or MSE, RMSE for regression).
- Compare model performance and discuss strengths and limitations of each approach.

3. Documentation

- Document each step clearly within the Jupyter Notebook, including code, methodologies, and rationale for choices.
- Use Markdown cells for explanations and code comments for clarity and readability.

4. Presentation

- Prepare a final presentation (10-15 slides) summarizing the project's objectives, methodology, results, and any conclusions or recommendations.
- Visual aids and code snippets in the presentation are encouraged to illustrate findings and model performances.

5. Deliverables

- Submit a complete Jupyter Notebook with all code and explanations, ensuring clear organization and documentation of the workflow.
- Submit a 10-15 slide presentation summarizing your project's objectives, methods, analysis, results, and key insights.

6. Submission Instructions

- Submit both the Jupyter Notebook and presentation through the Project Report Repository on Moodle.
- Ensure both partners' names are on the submission, and name the files appropriately (e.g., `ML_Project_Lastname1_Lastname2.ipynb`).
- Submissions are due by December 20, 2024, at 23:59. Late submissions may incur penalties.

7. Evaluation Criteria

Projects will be evaluated on:

- **Data Processing and EDA**: Thorough data cleaning, preprocessing, and insightful EDA.
- **Model Selection and Justification**: Appropriate choice of models and clear rationale.
- **Code Quality and Documentation**: Code that is well-written, efficient, and documented for readability.
- **Model Evaluation and Analysis**: Effective use of evaluation metrics and thoughtful analysis.
- **Final Presentation**: A clear, organized report and presentation, capturing the project's objectives, methods, and findings.